Business Case: Aerofit - Descriptive Statistics & Probability

1.**Introduction**

What is Aerofit ?

Aerofit is a leading brand in the field of fitness equipment. Aerofit provides a product range including machines such as treadmills, exercise bikes, gym equipment, and fitness accessories to cater to the needs of all categories of people.

Business Problem

The market research team at AeroFit wants to identify the characteristics of the target audience for each type of treadmill offered by the company, to provide a better recommendation of the treadmills to the new customers. The team decides to investigate whether there are differences across the product with respect to customer characteristics.

Objective

Create comprehensive customer profiles for each AeroFit treadmill product through descriptive analytics. Develop two-way contingency tables and analyze conditional and marginal probabilities to discern customer characteristics, facilitating improved product recommendations and informed business decisions.

2. Exploratory Data Analysis

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
import copy
```

```
# loading the dataset
df = pd.read_csv('aerofit_treadmill.txt')
```

```
df.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

```
df.shape
```

```
(180, 9)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   Product        180 non-null    object
 1   Age            180 non-null    int64 
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64 
 4   MaritalStatus  180 non-null    object
```

```
 5   Usage          180 non-null    int64
 6   Fitness        180 non-null    int64
 7   Income         180 non-null    int64
 8   Miles          180 non-null    int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

```
df.isnull().sum()
```

```
Product          0
Age              0
Gender           0
Education        0
MaritalStatus    0
Usage            0
Fitness          0
Income           0
Miles            0
dtype: int64
```

```
df['Product'].unique()
```

```
array(['KP281', 'KP481', 'KP781'], dtype=object)
```

```
df.describe(include="all")
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
count	180	180.000000	180	180.000000	180	180.000000	180.000000	180.000000	180.000000
unique	3	NaN	2	NaN	2	NaN	NaN	NaN	NaN
top	KP281	NaN	Male	NaN	Partnered	NaN	NaN	NaN	NaN
freq	80	NaN	104	NaN	107	NaN	NaN	NaN	NaN
mean	NaN	28.788889	NaN	15.572222	NaN	3.455556	3.311111	53719.577778	103.194444
std	NaN	6.943498	NaN	1.617055	NaN	1.084797	0.958869	16506.684226	51.863605
min	NaN	18.000000	NaN	12.000000	NaN	2.000000	1.000000	29562.000000	21.000000
25%	NaN	24.000000	NaN	14.000000	NaN	3.000000	3.000000	44058.750000	66.000000
50%	NaN	26.000000	NaN	16.000000	NaN	3.000000	3.000000	50596.500000	94.000000
75%	NaN	33.000000	NaN	16.000000	NaN	4.000000	4.000000	58668.000000	114.750000
max	NaN	50.000000	NaN	21.000000	NaN	7.000000	5.000000	104581.000000	360.000000

Observations:

- There are no missing values in the data.
- There are 3 unique products in the dataset.
- KP281 is the most frequent product.
- Minimum & Maximum age of the person is 18 & 50, mean is 28.79 and 75% of persons have age less than or equal to 33.
- Most of the people are having 16 years of education i.e. 75% of persons are having education <= 16 years.
- Out of 180 data points, 104's gender is Male and rest are the female.
- Standard deviation for Income & Miles is very high. These variables might have the outliers in it.

3.Univariate Analysis

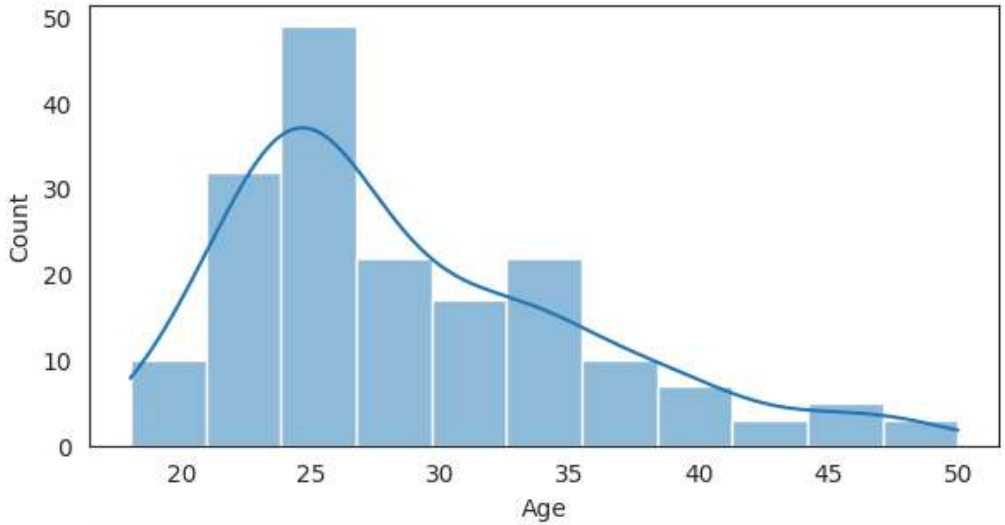
Understanding the distribution of numeric attributes:

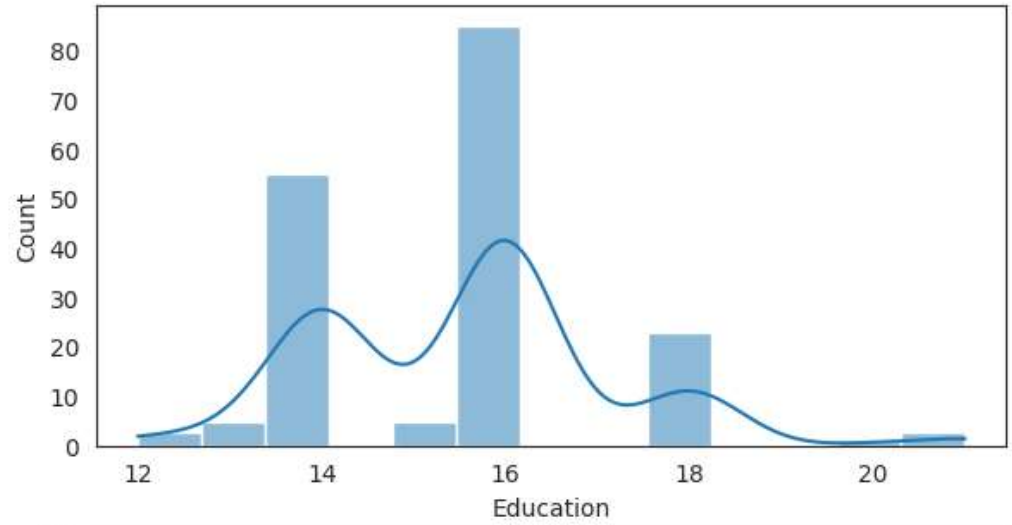
1. Age

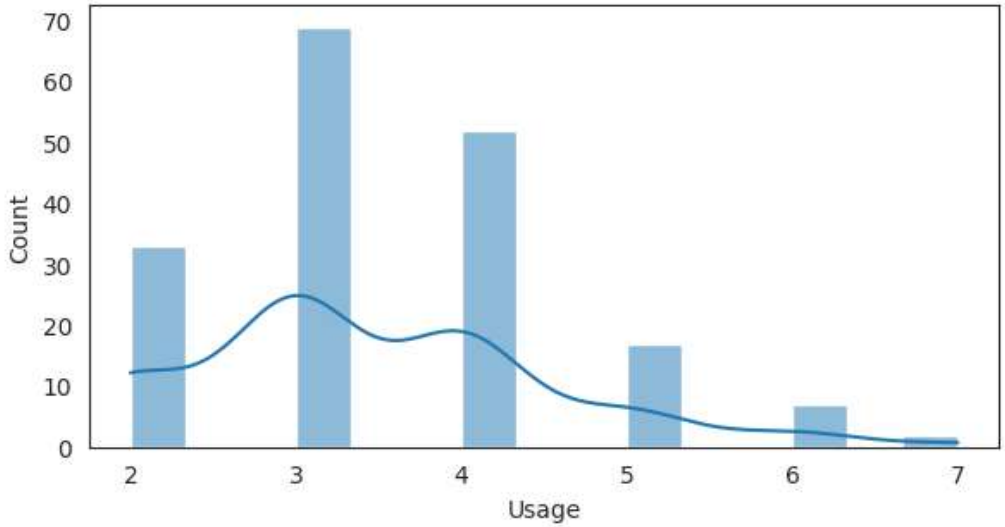
2. Education
3. Usage
4. Fitness
5. Income
6. Miles

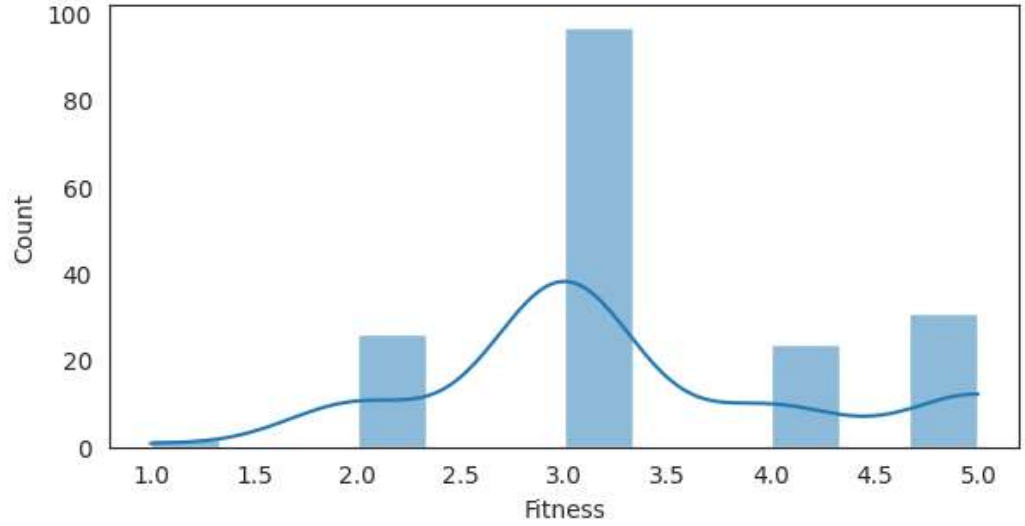
```
fig, axis = plt.subplots(nrows=3, ncols=2, figsize=(15,8))
fig.subplots_adjust(top=1.2)

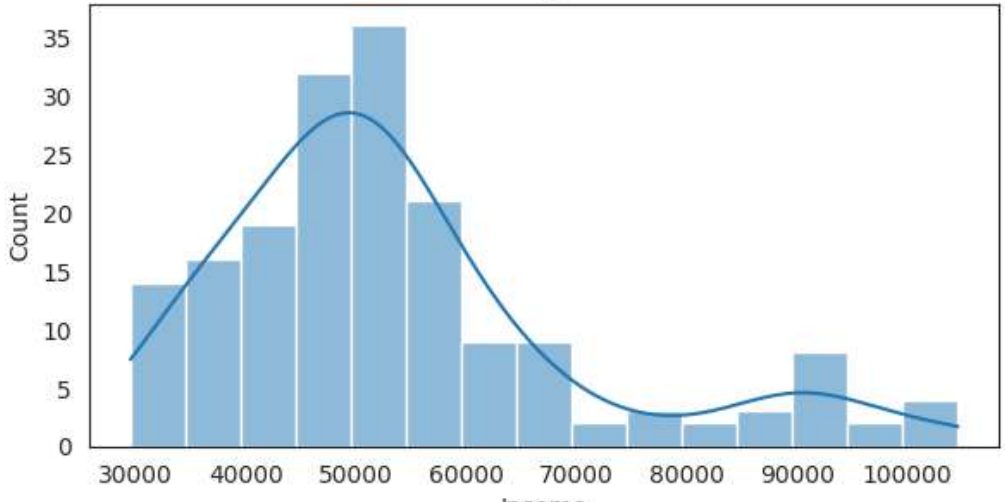
sns.histplot(data=df, x="Age", kde=True, ax=axis[0,0])
sns.histplot(data=df, x="Education", kde=True, ax=axis[0,1])
sns.histplot(data=df, x="Usage", kde=True, ax=axis[1,0])
sns.histplot(data=df, x="Fitness", kde=True, ax=axis[1,1])
sns.histplot(data=df, x="Income", kde=True, ax=axis[2,0])
sns.histplot(data=df, x="Miles", kde=True, ax=axis[2,1])
plt.show()
```

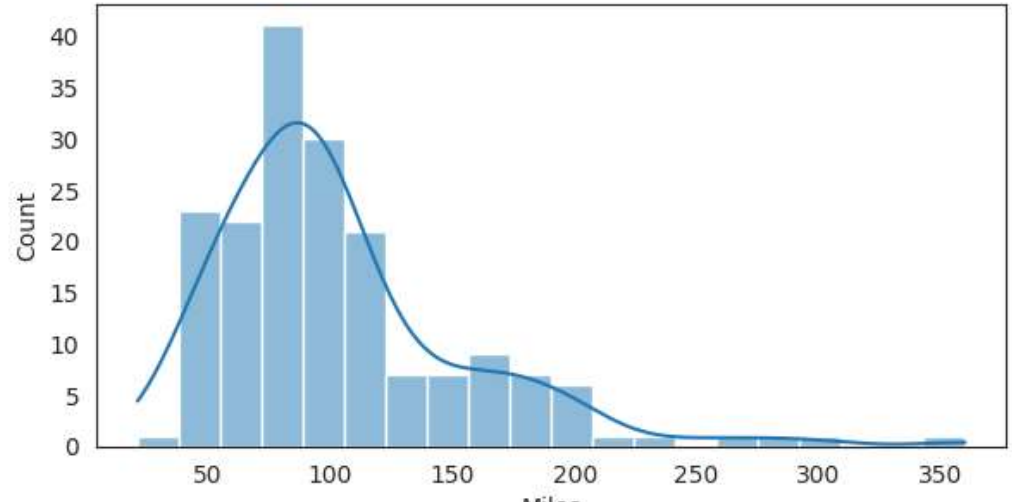

Count
Age


Count
Education


Count
Usage


Count
Fitness


Count


Count

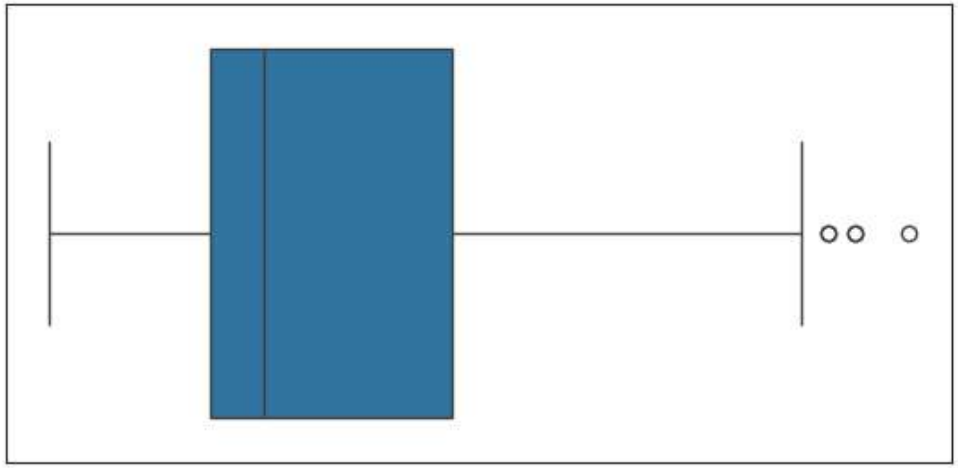
Income

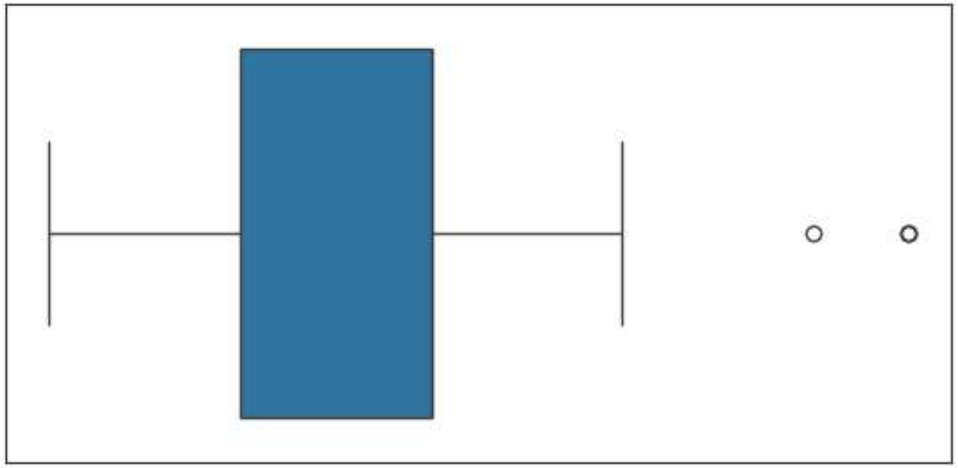
Miles

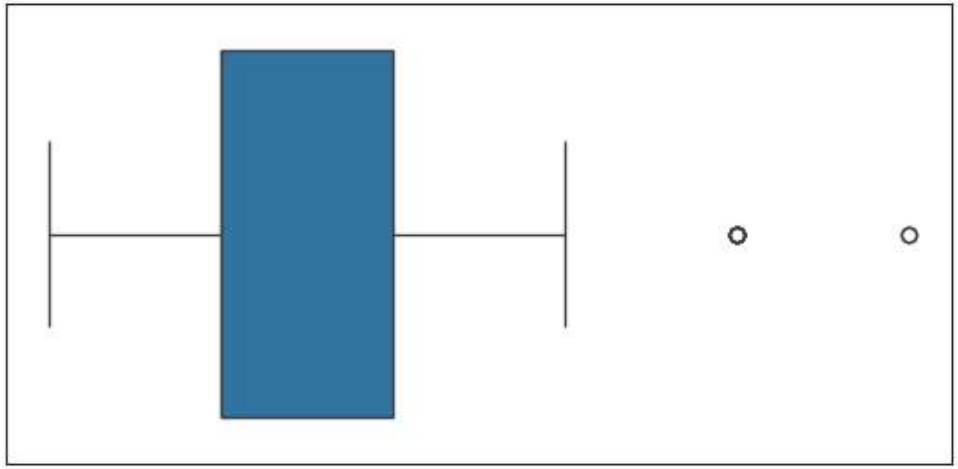
Outliers detection using BoxPlots

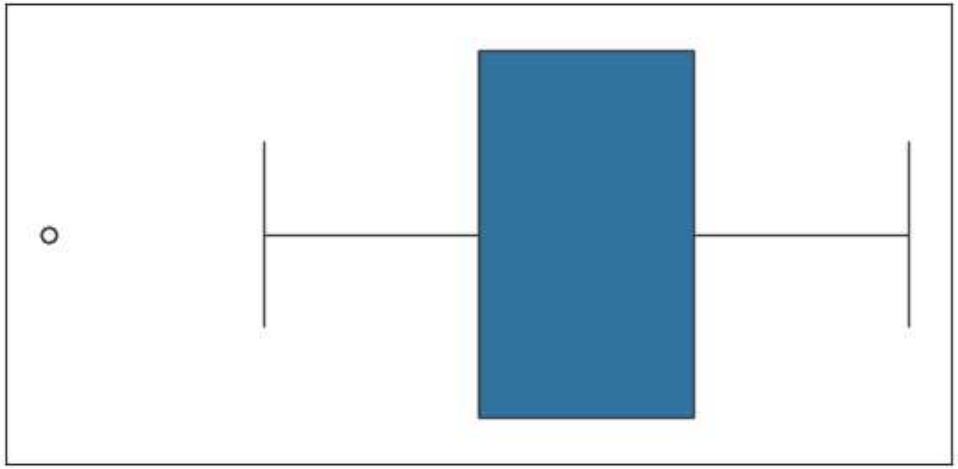
```
fig, axis = plt.subplots(nrows=3, ncols=2, figsize=(15, 8))
fig.subplots_adjust(top=1.2)

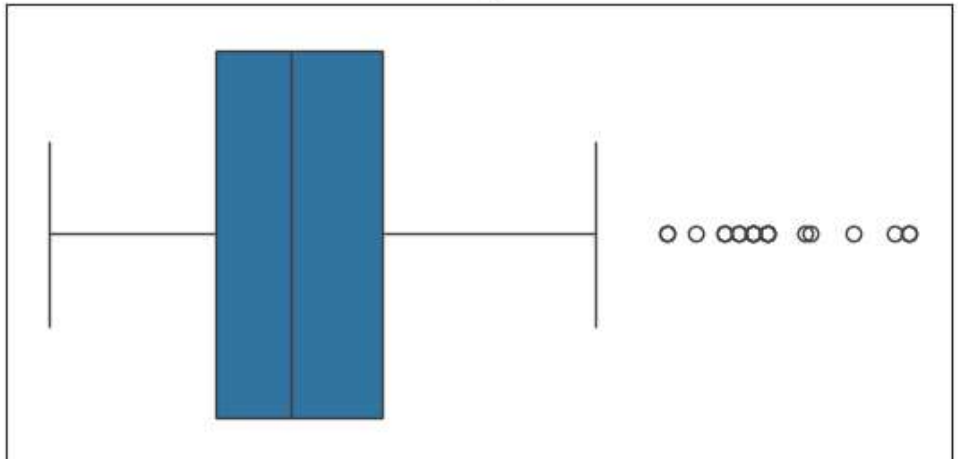
sns.boxplot(data=df, x="Age", orient='h', ax=axis[0,0])
sns.boxplot(data=df, x="Education", orient='h', ax=axis[0,1])
sns.boxplot(data=df, x="Usage", orient='h', ax=axis[1,0])
sns.boxplot(data=df, x="Fitness", orient='h', ax=axis[1,1])
sns.boxplot(data=df, x="Income", orient='h', ax=axis[2,0])
sns.boxplot(data=df, x="Miles", orient='h', ax=axis[2,1])
plt.show()
```

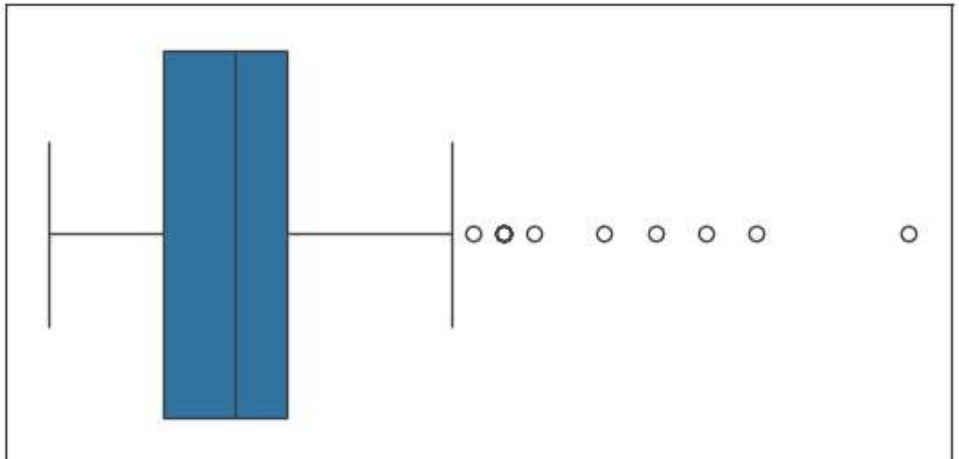

20
25
30
35
40
45
50
Age

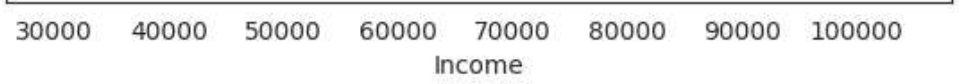

12
14
16
18
20
Education


2
3
4
5
6
7
Usage


1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Fitness









Obervation

Even from the boxplots it is quite clear that:

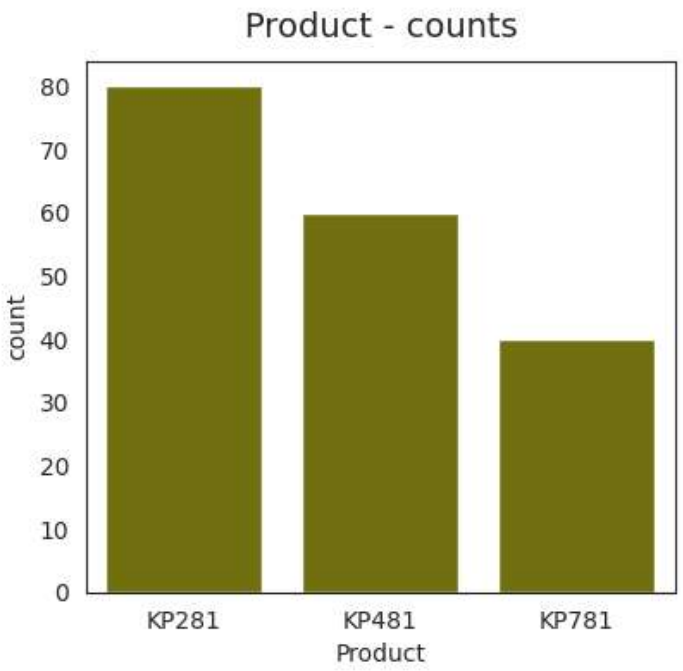
- Age, Education and Usage are having very few outliers.
- While Income and Miles are having more outliers.

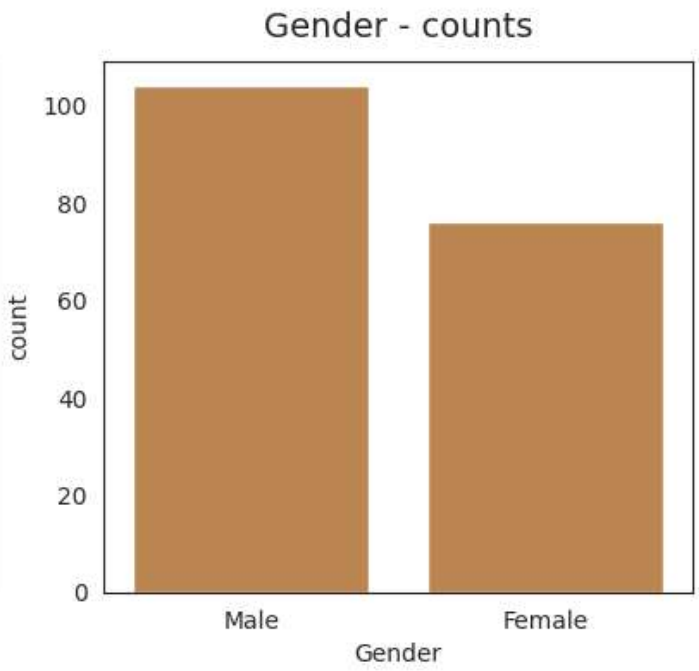
Understanding the distribution of the data for the qualitative attributes:

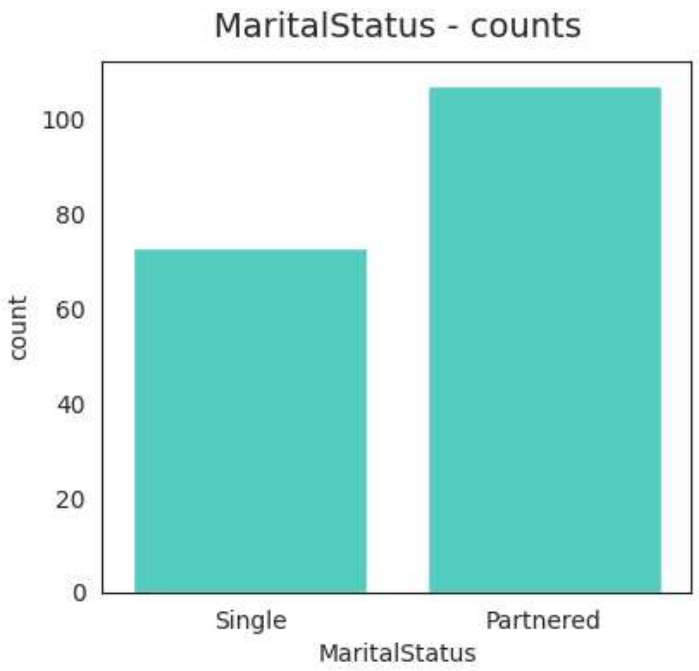
1. Product
2. Gender
3. MaritalStatus

```
fig, axs = plt.subplots(nrows=1, ncols=3, figsize=(15, 4))
sns.countplot(data=df, x='Product', ax=axs[0],color='olive')
sns.countplot(data=df, x='Gender', ax=axs[1],color='peru')
sns.countplot(data=df, x='MaritalStatus', ax=axs[2],color='turquoise')

axs[0].set_title("Product - counts", pad=10, fontsize=14)
axs[1].set_title("Gender - counts", pad=10, fontsize=14)
axs[2].set_title("MaritalStatus - counts", pad=10, fontsize=14)
plt.show()
```



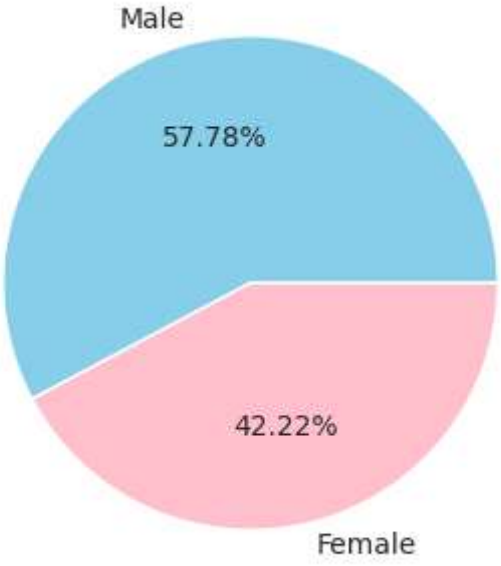




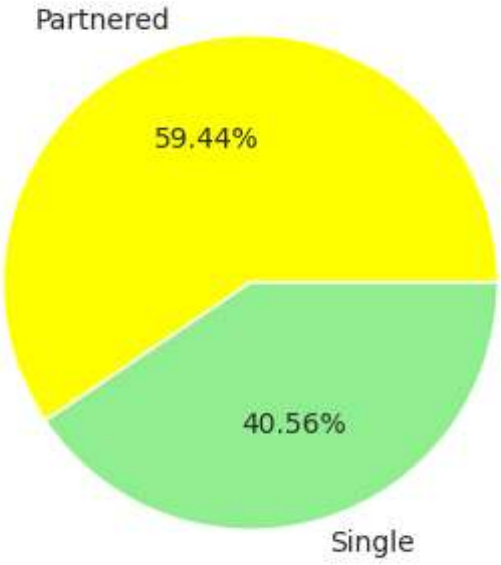
Obervations

- KP281 is the most frequent product.
- Thare are more Males in the data than Females.
- More Partnered persons are there in the data.

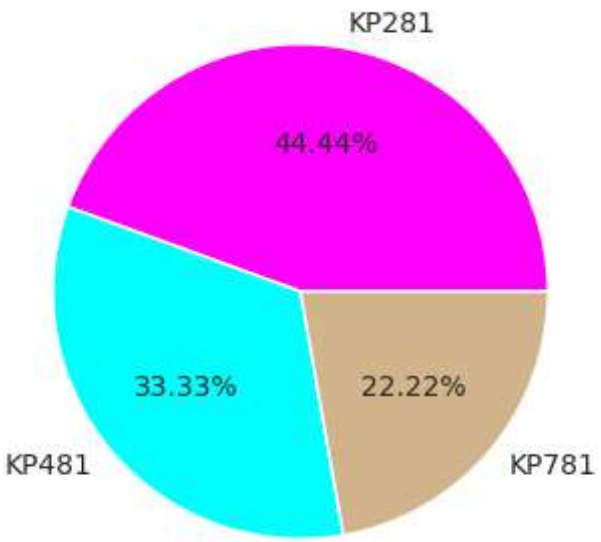
```
plt.figure(figsize=(4,4))
plt.pie(df['Gender'].value_counts().values,labels = df['Gender'].value_counts().index,autopct = '%.2f%%',colors=['skyblue', 'pink' ])
plt.show()
```




```
plt.figure(figsize=(4,4))
plt.pie(df['MaritalStatus'].value_counts().values,labels = df['MaritalStatus'].value_counts().index,autopct = '%.2f%%',colors=['yellow', 'ligl
plt.show()
```



```
plt.figure(figsize=(4,4))
plt.pie(df['Product'].value_counts().values,labels = df['Product'].value_counts().index,autopct = '%.2f%%',colors=['magenta', 'cyan','tan'])
plt.show()
```



To be precise - normalized count for each variable is shown below

```
df1 = df[['Product', 'Gender', 'MaritalStatus']].melt()
df1.groupby(['variable', 'value'])[['value']].count() / len(df)
```

		value
variable	value	
Gender	**Female**	0.422222
	Male	0.577778
MaritalStatus	**Partnered**	0.594444
	Single	0.405556
Product	**KP281**	0.444444
	KP481	0.333333
	KP781	0.222222

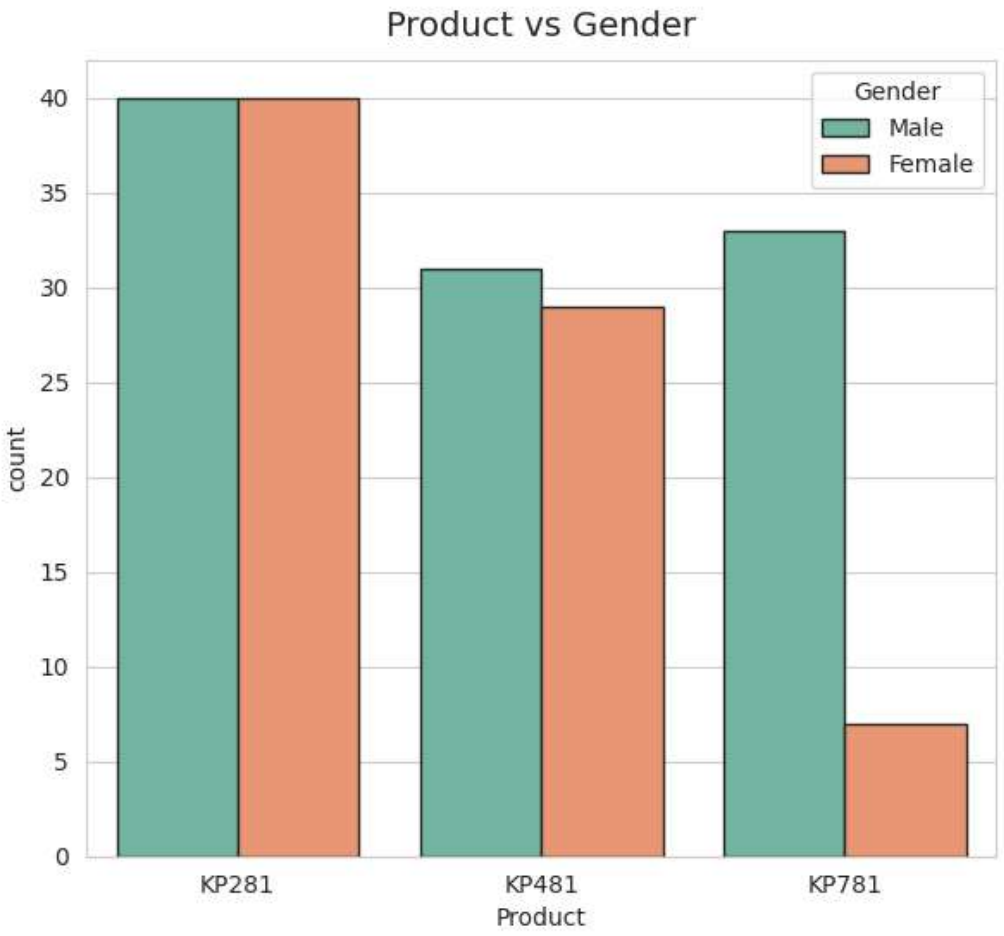
Obervations

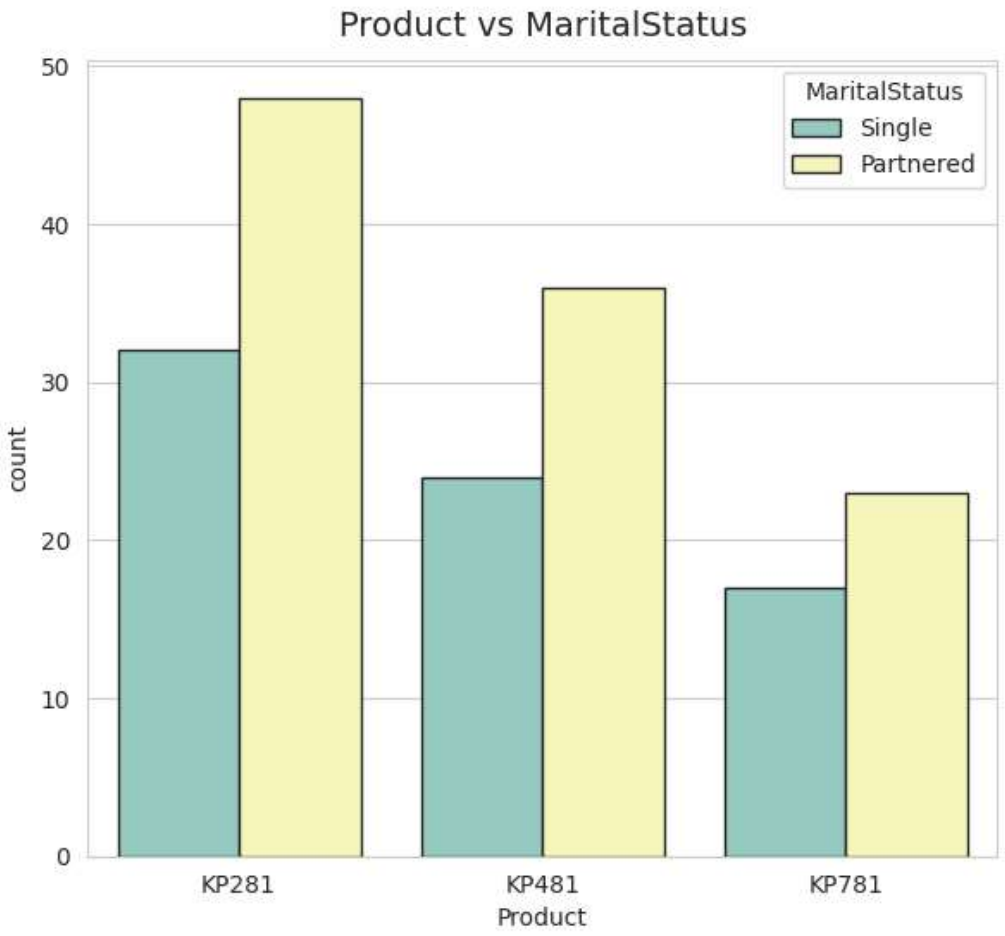
- **Product**
  - 44.44% of the customers have purchased **KP2821** product.
  - 33.33% of the customers have purchased **KP481** product.
  - 22.22% of the customers have purchased **KP781** product.
- **Gender**
  - 57.78% of the customers are **Male**.
- **MaritalStatus**
  - 59.44% of the customers are **Partnered**.

4.Bivariate Analysis

Checking if features : Gender or MaritalStatus have any effect on the product purchased.

```
sns.set_style(style='whitegrid')
fig, axs = plt.subplots(nrows=1, ncols=2, figsize=(15, 6))
sns.countplot(data=df, x='Product', hue='Gender', edgecolor="0.15", palette='Set2', ax=axs[0])
sns.countplot(data=df, x='Product', hue='MaritalStatus', edgecolor="0.15", palette='Set3', ax=axs[1])
axs[0].set_title("Product vs Gender", pad=10, fontsize=14)
axs[1].set_title("Product vs MaritalStatus", pad=10, fontsize=14)
plt.show()
```





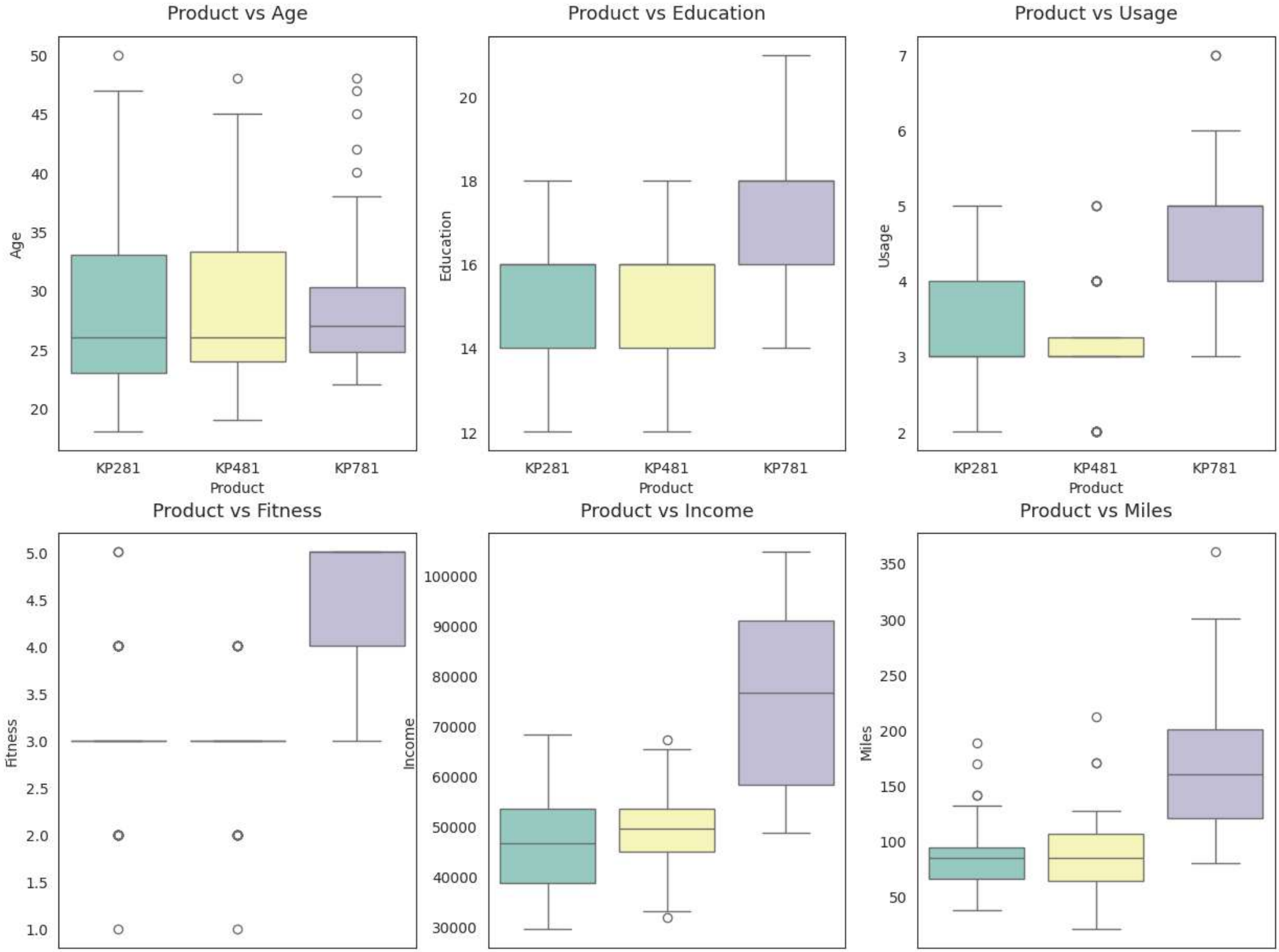
Obervations

- `Product vs Gender`
  - Equal number of males and females have purchased **KP281** product and Almost same for the product **KP481**
  - Most of the `Male` customers have purchased the **KP781** product.
- `Product vs MaritalStatus`
  - Customer who is `Partnered`, is more likely to purchase the product.

Checking if following features have any effect on the product purchased:

1. Age
2. Education
3. Usage
4. Fitness
5. Income
6. Miles

```
attrs = ['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']
sns.set_style("white")
fig, axs = plt.subplots(nrows=2, ncols=3, figsize=(15,8))
fig.subplots_adjust(top=1.2)
count = 0
for i in range(2):
    for j in range(3):
        sns.boxplot(data=df, x='Product', y=attrs[count], ax=axs[i,j], palette='Set3')
        axs[i,j].set_title(f"Product vs {attrs[count]}", pad=12, fontsize=13)
        count += 1
```


Product vs Age
Age
Product
KP281
KP481
KP781
Product vs Education
Education
Product vs Usage
Usage
Product vs Fitness
Fitness
Product vs Income
Income
Product vs Miles
Miles

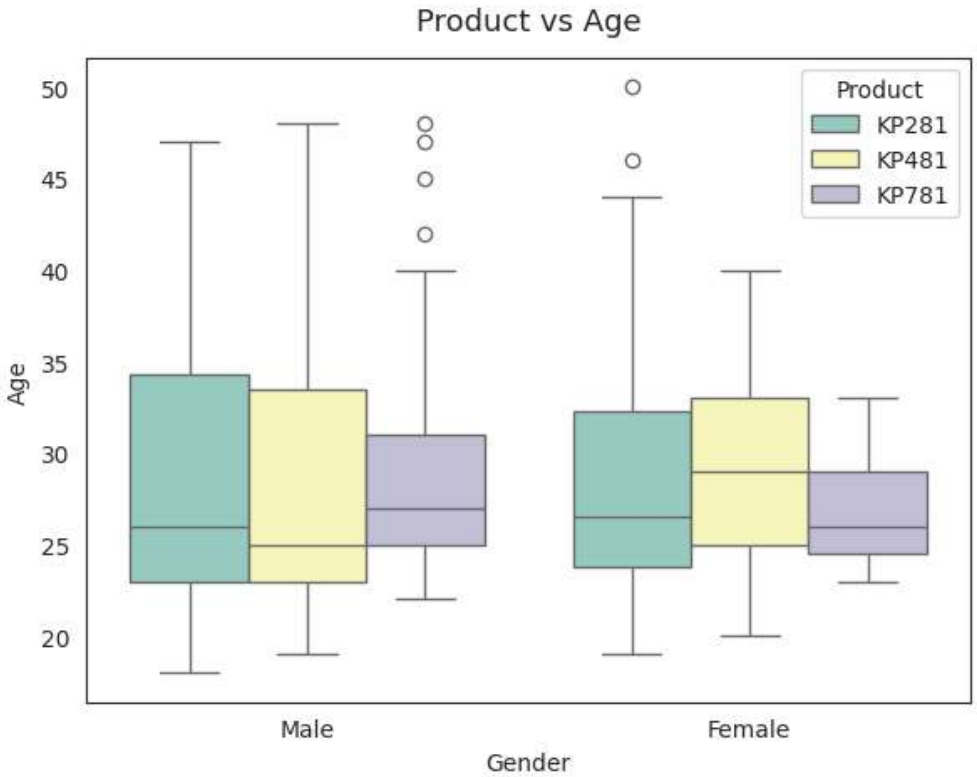


Observations

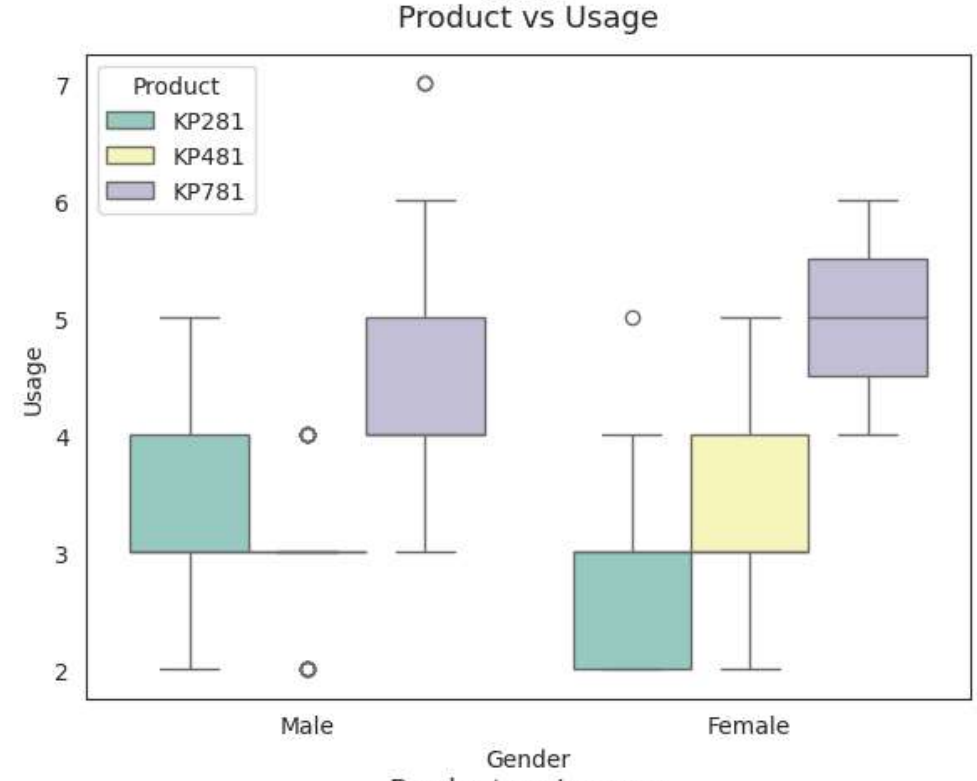
1. **`Product vs Age`**
   - Customers purchasing products `KP281 & KP481` are having same `Age` median value.
   - Customers whose age lies between 25-30, are more likely to buy `KP781` product
2. **`Product vs Education`**
   - Customers whose `Education` is greater than 16, have more chances to purchase the `KP781` product.
   - While the customers with `Education` less than 16 have equal chances of purchasing `KP281` or `KP481`.
3. **`Product vs Usage`**
   - Customers who are planning to use the treadmill greater than 4 times a week, are more likely to purchase the `KP781` product.
   - While the other customers are likely to purchasing `KP281` or `KP481`.
4. **`Product vs Fitness`**
   - The more the customer is fit (fitness >= 3), higher the chances of the customer to purchase the `KP781` product.
5. **`Product vs Income`**
   - Higher the `Income` of the customer (Income >= 60000), higher the chances of the customer to purchase the `KP781` product.
6. **`Product vs Miles`**
   - If the customer expects to walk/run greater than 120 Miles per week, it is more likely that the customer will buy `KP781` product.

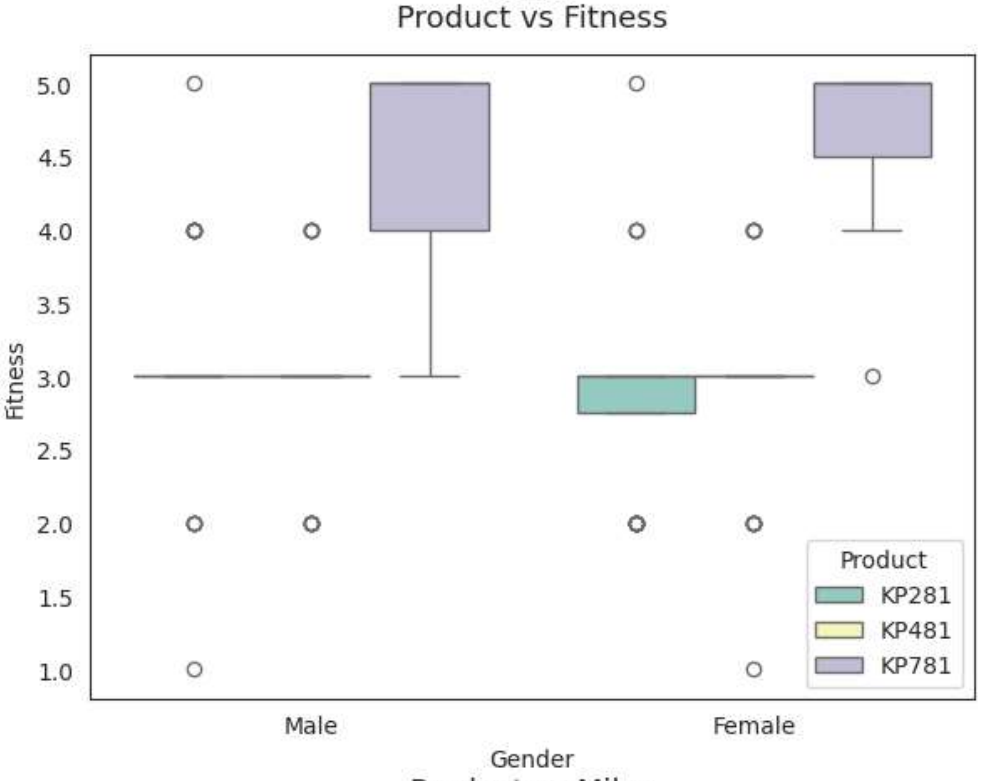
5.Multivariate Analysis

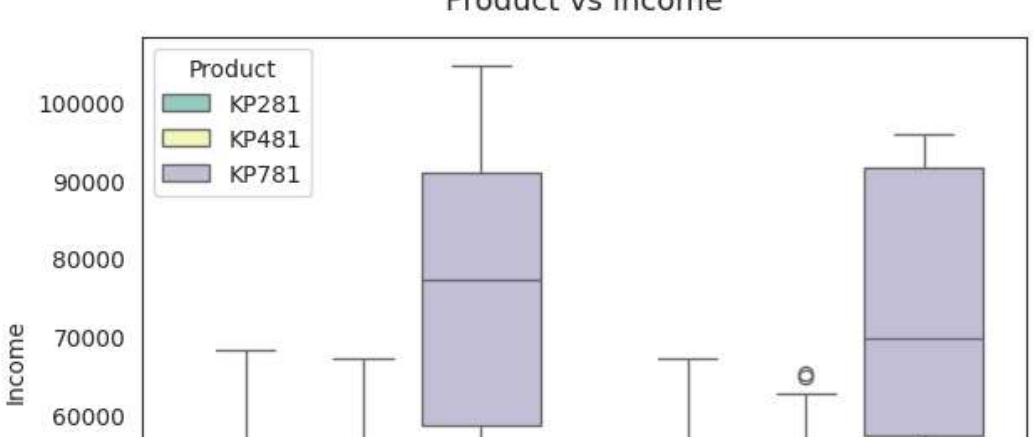
```
attrs = ['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']
sns.set_style("white")
fig, axs = plt.subplots(nrows=3, ncols=2, figsize=(15,11))
fig.subplots_adjust(top=1.3)
count = 0
for i in range(3):
    for j in range(2):
        sns.boxplot(data=df, x='Gender', y=attrs[count], hue='Product', ax=axs[i,j], palette='Set3')
        axs[i,j].set_title(f"Product vs {attrs[count]}", pad=12, fontsize=13)
        count += 1
```

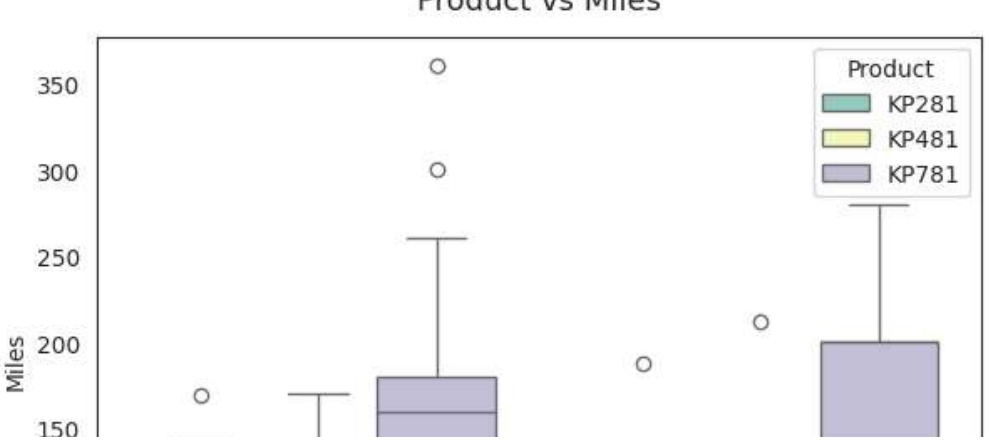

Product vs Age
Product
KP281
KP481
KP781
Age
50
45
40
35
30
25
20
Male
Female
Gender


Product vs Education
Product
KP281
KP481
KP781
Education
20
18
16
14
12
Male
Female
Gender


Product vs Usage
Product
KP281
KP481
KP781
Usage
7
6
5
4
3
2
Male
Female
Gender


Product vs Fitness
Product
KP281
KP481
KP781
Fitness
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Male
Female
Gender





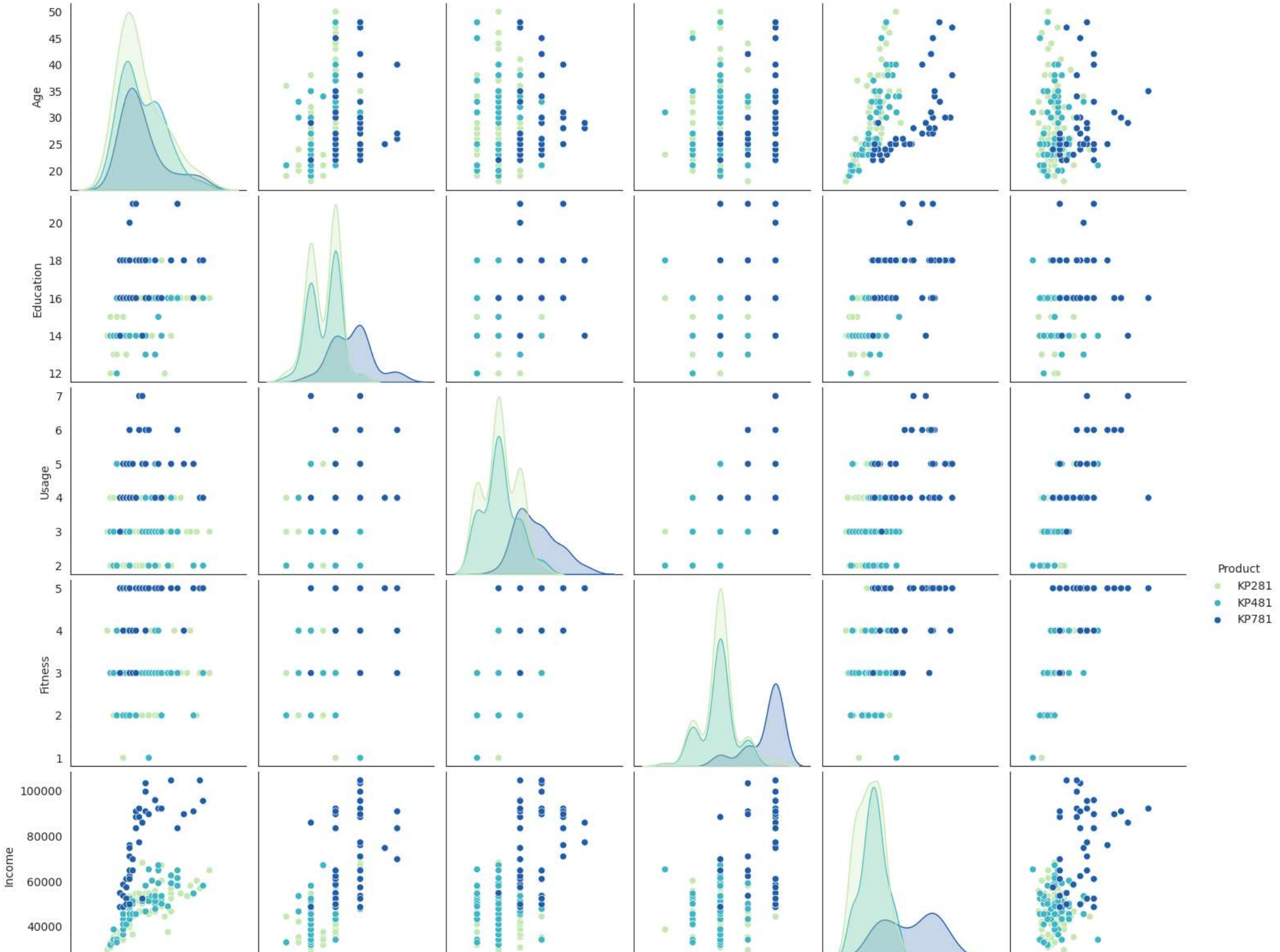
Obervations

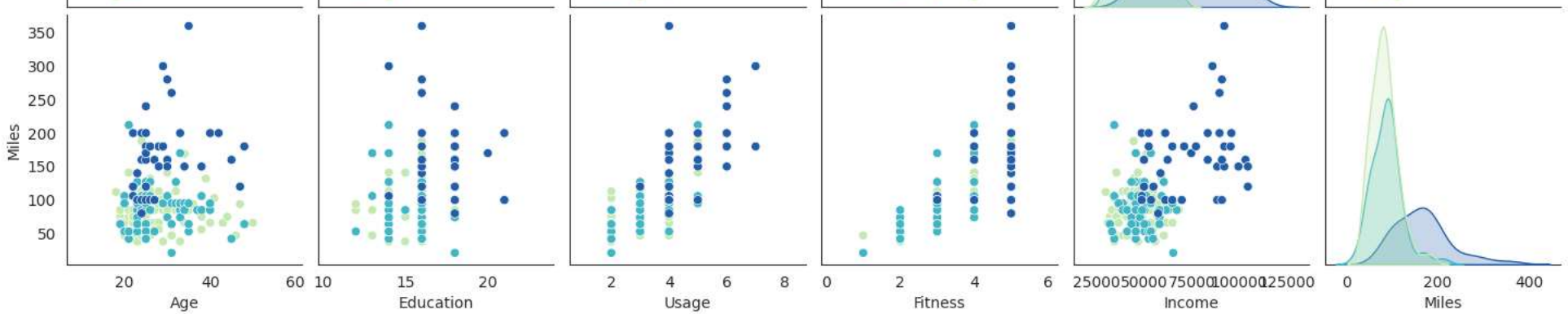
- Females planning to use treadmill 3-4 times a week, are more likely to buy `KP481` product

Pairplot

```
df_copy = copy.deepcopy(df)
```

```
sns.pairplot(df_copy, hue ='Product', palette= 'YlGnBu')
plt.show()
```


Age
Education
Usage
Fitness
Income
Product
KP281
KP481
KP781




```
df_copy.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   Product        180 non-null    object
 1   Age            180 non-null    int64 
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64 
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64 
 6   Fitness        180 non-null    int64 
 7   Income         180 non-null    int64 
 8   Miles          180 non-null    int64 
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

6. Computing Probability - Marginal, Conditional Probability

6.1 Probability of product purchase with respect to gender

```
pd.crosstab(index =df['Product'],columns = df['Gender'],margins = True,normalize = True ).round(2)
```

Gender	Female	Male	All
Product			
KP281	0.22	0.22	0.44
KP481	0.16	0.17	0.33
KP781	0.04	0.18	0.22
All	0.42	0.58	1.00

Insights

1. The **Probability** of a treadmill being purchased by a **female is 42%**.
    - **The conditional probability** of purchasing the treadmill model given that the customer is **female** is
        - For Treadmill model KP281 - **22%**
        - For Treadmill model KP481 - **16%**
        - For Treadmill model KP781 - **4%**
2. The **Probability** of a treadmill being purchased by a **male is 58%**.
    - **The conditional probability** of purchasing the treadmill model given that the customer is **male** is -
        - For Treadmill model KP281 - **22%**
        - For Treadmill model KP481 - **17%**
        - For Treadmill model KP781 - **18%**

6.2 Probability of product purchase w.r.t. Age

```
pd.crosstab(index =df['Product'],columns = df['Age'],margins = True,normalize = True ).round(2)
```

Age	18	19	20	21	22	23	24	25	26	27	...	41	42	43	44	45	46	47	48	50	All
Product																					
KP281	0.01	0.02	0.01	0.02	0.02	0.04	0.03	0.04	0.04	0.02	...	0.01	0.00	0.01	0.01	0.00	0.01	0.01	0.00	0.01	0.44
KP481	0.00	0.01	0.02	0.02	0.00	0.04	0.02	0.06	0.02	0.01	...	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.33
KP781	0.00	0.00	0.00	0.00	0.02	0.02	0.02	0.04	0.01	0.02	...	0.00	0.01	0.00	0.00	0.01	0.00	0.01	0.01	0.00	0.22
All	0.01	0.02	0.03	0.04	0.04	0.10	0.07	0.14	0.07	0.04	...	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1.00

4 rows × 33 columns

NOTE:From the above we can see that it is difficult to calculate probability among different age groups so lets create some bins for different age groups according to the range of age values and analyze.Similarly for other colums as well

Adding new columns for better analysis

- Creating New Column and Categorizing values in Age,Education,Income and Miles to different classes for better visualization

1. Young Adult: from 18 - 25
2. Adults: from 26 - 35
3. Middle Aged Adults: 36-45
4. Elder :46 and above

```
bin_range1 = [17,25,35,45,float('inf')]
bin_labels1 = ['Young Adults', 'Adults', 'Middle Aged', 'Elder']
df_copy['Age_group'] = pd.cut(df_copy['Age'],bins = bin_range1,labels = bin_labels1)
```

```
pd.crosstab(index =df_copy['Product'],columns = df_copy['Age_group'],margins = True,normalize = True ).round(2)
```

Age_group	Young Adults	Adults	Middle Aged	Elder	All
Product					
KP281	0.19	0.18	0.06	0.02	0.44
KP481	0.16	0.13	0.04	0.01	0.33
KP781	0.09	0.09	0.02	0.01	0.22
All	0.44	0.41	0.12	0.03	1.00

Insights

1. The **Probability** of a treadmill being purchased by a `Young Adult(18-25) is 44%`.
    - **The conditional probability** of purchasing the treadmill model given that the customer is `Young Adult` is
        - For Treadmill model KP281 - **19%**
        - For Treadmill model KP481 - **16%**
        - For Treadmill model KP781 - **9%**
2. The **Probability** of a treadmill being purchased by a `Adult(26-35) is 41%`.
    - **The conditional probability** of purchasing the treadmill model given that the customer is `Adult` is -
        - For Treadmill model KP281 - **18%**
        - For Treadmill model KP481 - **13%**
        - For Treadmill model KP781 - **9%**
3. The **Probability** of a treadmill being purchased by a **Middle Aged(36-45) is 12%**.
4. The **Probability** of a treadmill being purchased by a **Elder(Above 45) is only 3%**.

6.3 Probability of product purchase w.r.t. Education level

Education Column

- Categorizing the values in education column in 3 different buckets:

1. Primary Education: upto 12
2. Secondary Education: 13 to 15
3. Higher Education: 16 and above

```
bin_range2 = [0,12,15,float('inf')]
bin_labels2 = ['Primary Education', 'Secondary Education', 'Higher Education']
df_copy['Edu_group'] = pd.cut(df_copy['Education'],bins = bin_range2,labels = bin_labels2)
```

```
pd.crosstab(index =df_copy['Product'],columns = df_copy['Edu_group'],margins = True,normalize = True ).round(2)
```

Edu_group	Primary Education	Secondary Education	Higher Education	All
Product				
KP281	0.01	0.21	0.23	0.44
KP481	0.01	0.14	0.18	0.33
KP781	0.00	0.01	0.21	0.22
All	0.02	0.36	0.62	1.00

Insights

1. The `Probability` of a treadmill being purchased by a customer with `Higher Education(Above 15 Years) is 62%`.
    - `The conditional probability` of purchasing the treadmill model given that the customer has `Higher Education` is
        - For Treadmill model KP281 - **23%**
        - For Treadmill model KP481 - **18%**
        - For Treadmill model KP781 - **21%**

2. The `Probability` of a treadmill being purchased by a customer with `Secondary Education(13-15 yrs) is 36%`.
    - `The conditional probability` of purchasing the treadmill model given that the customer has `Secondary Education` is -
        - For Treadmill model KP281 - **21%**
        - For Treadmill model KP481 - **14%**
        - For Treadmill model KP781 - **1%**
3. The `Probability` of a treadmill being purchased by a customer with `Primary Education(0 to 12 yrs) is only 2%`.

6.4 Probability of product purchase w.r.t. Income

Income Column

- Categorizing the values in Income column in 4 different buckets:

1. Low Income - Upto 40,000
2. Moderate Income - 40,000 to 60,000
3. High Income - 60,000 to 80,000
4. Very High Income - Above 80,000

```
bin_range3 = [0,40000,60000,80000,float('inf')]
bin_labels3 = ['Low Income','Moderate Income','High Income','Very High Income']
df_copy['Income_group'] = pd.cut(df_copy['Income'],bins = bin_range3,labels = bin_labels3)
```

```
pd.crosstab(index =df_copy['Product'],columns = df_copy['Income_group'],margins = True,normalize = True ).round(2)
```

Income_group	Low Income	Moderate Income	High Income	Very High Income	All
Product					
KP281	0.13	0.28	0.03	0.00	0.44
KP481	0.05	0.24	0.04	0.00	0.33
KP781	0.00	0.06	0.06	0.11	0.22
All	0.18	0.59	0.13	0.11	1.00

Insights

1. The `Probability` of a treadmill being purchased by a customer with `Low Income(<40k) is 18%`.
    - `The conditional probability` of purchasing the treadmill model given that the customer has `Low Income` is -
        - For Treadmill model KP281 - **13%**
        - For Treadmill model KP481 - **5%**
        - For Treadmill model KP781 - **0%**
2. The `Probability` of a treadmill being purchased by a customer with `Moderate Income(40k - 60k) is 59%`.
    - `The conditional probability` of purchasing the treadmill model given that the customer has `Moderate Income` is -
        - For Treadmill model KP281 - **28%**
        - For Treadmill model KP481 - **24%**
        - For Treadmill model KP781 - **6%**
3. The `Probability` of a treadmill being purchased by a customer with `High Income(60k - 80k) is 13%`
    - `The conditional probability` of purchasing the treadmill model given that the customer has `High Income` is -
        - For Treadmill model KP281 - **3%**
        - For Treadmill model KP481 - **4%**

    - For Treadmill model KP781 - **6%**

4. The `Probability` of a treadmill being purchased by a customer with `Very High Income(>80k) is 11%`
    - `The conditional probability` of purchasing the treadmill model given that the customer has `High Income` is -
        - For Treadmill model KP281 - **0%**
        - For Treadmill model KP481 - **0%**

6.5 Probability of product purchase w.r.t. weekly mileage

Miles column

- Categorizing the values in miles column in 4 different buckets:

1. Light Activity - Upto 50 miles
2. Moderate Activity - 51 to 100 miles
3. Active Lifestyle - 101 to 200 miles
4. Fitness Enthusiast - Above 200 miles

```
bin_range4 = [0,50,100,200,float('inf')]
bin_labels4 = ['Light Activity', 'Moderate Activity', 'Active Lifestyle', 'Fitness Enthusiast ']
df_copy['Miles_group'] = pd.cut(df_copy['Miles'],bins = bin_range4,labels = bin_labels4)
```

```
pd.crosstab(index =df_copy['Product'],columns = df_copy['Miles_group'],margins = True,normalize = True ).round(2)
```

Miles_group Product	Light Activity	Moderate Activity	Active Lifestyle	Fitness Enthusiast	All
KP281	0.07	0.28	0.10	0.00	0.44
KP481	0.03	0.22	0.08	0.01	0.33
KP781	0.00	0.04	0.15	0.03	0.22
All	0.09	0.54	0.33	0.03	1.00

Insights

1. The `Probability` of a treadmill being purchased by a customer with lifestyle of `Light Activity(0 to 50 miles/week) is 9%`.
    - `The conditional probability` of purchasing the treadmill model given that the customer has `Light Activity Lifestyle` is -
        - For Treadmill model KP281 - **7%**
        - For Treadmill model KP481 - **3%**
        - For Treadmill model KP781 - **0%**
2. The `Probability` of a treadmill being purchased by a customer with lifestyle of `Moderate Activity(51 to 100 miles/week) is 54%`.
    - `The conditional probability` of purchasing the treadmill model given that the customer with lifestyle of `Moderate Activity` is -
        - For Treadmill model KP281 - **28%**
        - For Treadmill model KP481 - **22%**
        - For Treadmill model KP781 - **4%**
2. The `Probability` of a treadmill being purchased by a customer has `Active Lifestyle(100 to 200 miles/week) is 33%`.
    - `The conditional probability` of purchasing the treadmill model given that the customer has `Active Lifestyle` is -
        - For Treadmill model KP281 - **10%**
        - For Treadmill model KP481 - **8%**
        - For Treadmill model KP781 - **15%**

4. The `Probability` of a treadmill being purchased by a customer who is `Fitness Enthusiast(>200 miles/week) is 3% only`

6.6 Probability of product purchase w.r.t. Marital Status

```
pd.crosstab(index =df['Product'],columns = df['MaritalStatus'],margins = True,normalize = True ).round(2)
```

MaritalStatus	Partnered	Single	All
Product			
KP281	0.27	0.18	0.44
KP481	0.20	0.13	0.33
KP781	0.13	0.09	0.22
All	0.59	0.41	1.00

Insights

1. The `Probability` of a treadmill being purchased by a `Married Customer is 59%`.
    - `The conditional probability` of purchasing the treadmill model given that the customer is `Married` is
        - For Treadmill model KP281 - **27%**
        - For Treadmill model KP481 - **20%**
        - For Treadmill model KP781 - **13%**
2. The `Probability` of a treadmill being purchased by a `Unmarried Customer is 41%`.
    - `The conditional probability` of purchasing the treadmill model given that the customer is `Unmarried` is -
        - For Treadmill model KP281 - **18%**
        - For Treadmill model KP481 - **13%**

- For Treadmill model KP781 - **9%**

6.7 Probability of product purchase w.r.t. Weekly Usage

```
pd.crosstab(index =df['Product'],columns = df['Usage'],margins = True,normalize = True ).round(2)
```

Usage	2	3	4	5	6	7	All